

the items and associate them with classes while scoring them when compared to the actual object.

The final function loop calls for the input of all the images one by one and specifies details such as the learning rate, the model used, the update algorithm and several other parameters that will affect the accuracy of the model.

You might have noticed that the code uses the sample COCO dataset from Google but can be changed to include other image datasets as well such as the Kitti dataset and the Open Images dataset.

The detection model used in the code is just one among the several that can be created and referenced. Others that often produce great results include the Single Shot Detectors(SSDs) and MobileNets models that offer great computational capacities.

The VTune™ Amplifier code shows how to streamline multiple projects to the system by using the Source Function / Function / Call Stack grouping method. This is an improvement to the default Function / Call Stack since TensorFlow was built using the Intel® Math Kernel Library for Deep Neural Network.

MOVING ON

Keep updates and saves to the model in a separate folder. The system's accuracy of object detection will improve once the algorithm has spanned through the images and runs through them again in a refined procedure. And once the output shows the various objects being present, you will see that the items have been differentiated with different percentages as to how close they match to the actual item specified at the input.

This is also a great time to create an online prompt system that takes images and shows you the percentage proximity to an object which is entered by the user. A simple one can be replicated by using the Android SDK on devices as well. And as always, consider using a framework that is optimised to run on whichever hardware you are using. We've already touched upon how much of a difference this can make at the beginning of the previous chapter.

In the next chapter we uncover the code and see what it looks like on an actual environment along with some tips and inferences to help the coding process feel simpler and exact. 